

Dry Run

Input Array: {3, 7, 2, 5}

$$\text{Total sum} = 3 + 7 + 2 + 5 = 17$$

Target Capacity ($S/2$): $17/2 = 8$ (Integer division)

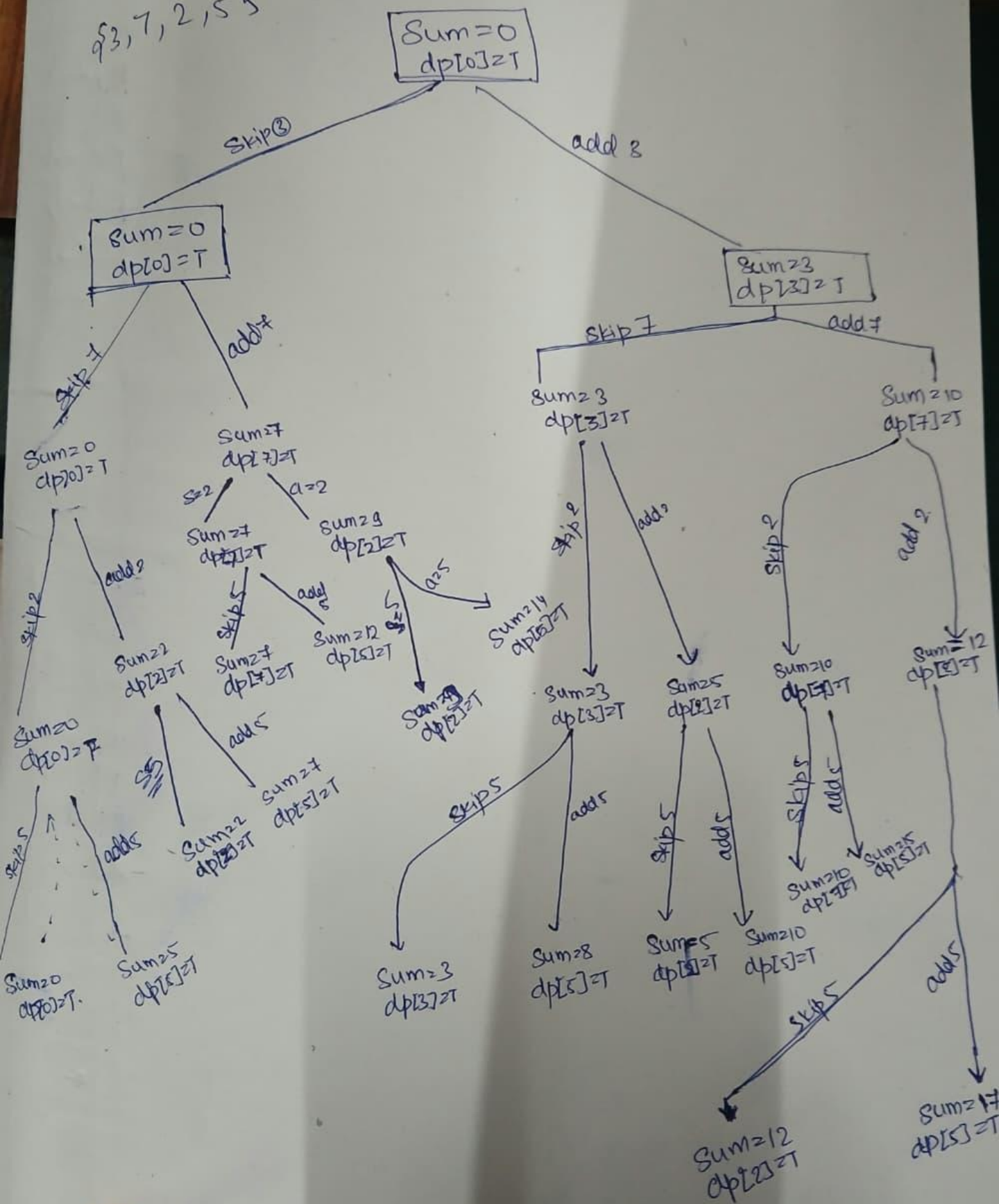
We initialize a DP array a size 9 (Indices 0 through 8).
 $dp[0]$ is initialized to T (True), all other as F (False).

Step	num	0	1	2	3	4	5	6	7	8
0	—	T	F	F	F	F	F	F	F	F
1	+ 3	T	F	F	T	F	F	F	F	F
2	+ 7	T	F	F	T	F	F	F	T	F
3	+ 2	T	F	T	T	F	T	F	T	F
4	+ 5	T	F	T	T	F	T	F	T	T

Annotations:

- Step 1: $dp[3] dp[0]$
- Step 2: $dp[7] dp[0]$
- Step 3: $dp[2] dp[0]$ and $dp[5] dp[0]$
- Step 4: $dp[8] dp[3]$ (marked with a star)

93, 7, 2, 53



Time & Space Complexity

Time Complexity: $O(n \times \text{target})$

↓

Simplifies to $O(n \times \text{totalsum})$

as we iterate through n elements, and for each element, we iterate up to $\frac{\text{totalsum}}{2}$ times in inner loop.

Space Complexity: $O(\text{totalsum})$

we use a 1D boolean array of size $\frac{\text{totalsum}}{2} + 1$. → optimization to traditional

2D, DP matrix...

⇒ $O(n \times \text{total sum})$ space.